

Weighted Worst-Group Reliability under Covariate Shift: A Theory-First Study of Hidden Deployment Failures

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April 17, 2026

Abstract

Models are often tested on one population and then deployed on another. After this shift, the model can still look reliable on average while failing badly on a subgroup. This paper studies that problem for binary prediction under covariate shift.

Our goal is simple. We want to estimate reliability on the target population using source data, and we want to understand when average metrics hide subgroup failures. Under covariate shift, target quantities can be written as importance-weighted source quantities. Using this idea, we define subgroup-aware reliability, worst-group reliability, and a hidden-failure gap that measures how much aggregate evaluation misses.

We then derive estimation guarantees and show a basic negative result: aggregate calibration can look perfect even when subgroup calibration is poor. We also study these ideas in controlled tabular experiments, where the same patterns appear in practice. The main message is that reliability under shift should be studied as a weighted subgroup problem, not as a single average score.

1 Introduction

A model is usually tested on one population and then deployed on another. That change in population can be enough to break reliability. A model may still look acceptable on average, but become unreliable on a subgroup that matters in practice.

Here is a simple example. Suppose a hospital model predicts the probability of complications after a procedure. On the validation set, patients from all departments are mixed together, and the model looks well calibrated overall. After deployment, the patient mix changes: one department becomes much more common, and that department is exactly where the model is overconfident. The overall score may still look decent, but the deployment problem is real.

This paper studies that situation in a clean theoretical setting. Here X is the feature vector, $Y \in \{0, 1\}$ is the label, and the model outputs a score

$$s(X) \in [0, 1].$$

If the model says 0.8, we would like that to mean “about 80% chance of being correct” or “about 80% chance of class 1.” In simple terms, calibration asks whether predicted probabilities match empirical frequencies.

For one bin of scores, an intuitive calibration gap is

$$\text{gap} \approx |\text{observed frequency} - \text{predicted probability}|.$$

Overall calibration is then some average of these gaps across bins. The problem is that this is still an average. If one subgroup is strongly overconfident and another is strongly underconfident, the two effects can cancel out.

We therefore focus on *worst-group reliability*. If $\mathcal{G}$ is a family of subgroups, the quantity of interest is roughly

$$\text{worst-group error} = \max_{g \in \mathcal{G}} \text{error on group } g.$$

We also look at the difference between worst-group error and overall error:

$$\text{hidden-failure gap} = \text{worst-group error} - \text{overall error}.$$

If this gap is large, then the aggregate metric is hiding something important.

Our second ingredient is distribution shift. We write P for the source distribution and Q for the target distribution. We work under the standard covariate-shift assumption

$$P(Y | X) = Q(Y | X), \quad P_X \neq Q_X.$$

In words: the mix of inputs changes, but the label mechanism given the input stays the same. In that case, target expectations can be expressed using source data and a weight

$$w(x) = \frac{dQ_X}{dP_X}(x).$$

So the basic rule is

$$\text{target average} \approx \frac{1}{n} \sum_{i=1}^n w(X_i) h(X_i, Y_i),$$

for an appropriate quantity h . This is the key mathematical idea in the paper. It says that target reliability is a weighted source problem.

Once we write the problem this way, three things become clear.

1. Target reliability can be estimated by weighted source averages.
2. Worst-group reliability is harder than aggregate reliability because rare groups and unstable weights reduce usable sample size.
3. Aggregate calibration can be misleading even when the target average looks good.

The rest of the paper makes this precise. We define subgroup-aware target reliability quantities, derive weighted transport identities, and prove estimation bounds that separate sampling error from density-ratio estimation error. We also give a simple construction showing that aggregate calibration can be perfect while subgroup calibration is poor.

The contribution of the paper is therefore simple. It is not a new software interface. It is a scientific viewpoint:

reliability under deployment shift should be analyzed as a weighted subgroup estimation problem, not as a single average metric.

The experiments are there to test this viewpoint in controlled settings, not to replace the theory. They show the same qualitative pattern throughout the paper: as shift grows, weighting becomes statistically harder, and subgroup failures become easier to miss if one only reports average metrics.

2 Problem Setting

2.1 Source and target distributions

Let P denote the source distribution and let Q denote the target distribution. Here $X \in \mathcal{X}$ is the feature vector and $Y \in \{0, 1\}$ is the binary label. We observe labeled source samples

$$(X_1, Y_1), \dots, (X_n, Y_n) \sim P,$$

and unlabeled or lightly labeled target covariates

$$X'_1, \dots, X'_m \sim Q_X.$$

We study a fixed score function $s : \mathcal{X} \rightarrow [0, 1]$. In examples, $s(X)$ is the model's predicted probability of class 1.

Assumption 1 (Covariate shift). The conditional label mechanism is unchanged:

$$P(Y | X) = Q(Y | X),$$

while the marginal feature distribution may change:

$$P_X \neq Q_X.$$

Assumption 2 (Overlap). The target covariate distribution is absolutely continuous with respect to the source covariate distribution, so the density ratio

$$w(x) = \frac{dQ_X}{dP_X}(x)$$

exists and is finite almost surely under P .

These are standard assumptions in the covariate-shift literature ([Sugiyama et al., 2007](#); [Park et al., 2020](#)). Their appeal is that target expectations can be written as weighted source expectations.

2.2 Groups, bins, and moment-based reliability

Let $\mathcal{G}$ be a finite family of subgroup indicators $g : \mathcal{X} \rightarrow \{0, 1\}$. In applications, each g may represent a categorical slice, a discretized numerical range, or an intersectional subgroup. Let

$$\Pi = \{I_1, \dots, I_B\}$$

be a fixed partition of the score range $[0, 1]$.

Example 1 (Simple subgroup family). If the application is hospital risk prediction, one may take

$$g_{\text{cardiology}}(x) = \mathbf{1}\{\text{department} = \text{cardiology}\}$$

or

$$g_{\text{high-age}}(x) = \mathbf{1}\{\text{age} \geq 75\}.$$

If the model score is binned into intervals such as $[0, 0.2)$, $[0.2, 0.4)$, and so on, then we can ask whether the model is well calibrated inside each bin and inside each subgroup.

For each bin I_b and subgroup indicator $g \in \mathcal{G}$, define the target calibration residual moment

$$\theta_{b,g}^Q = \mathbb{E}_Q[(Y - s(X))\mathbf{1}\{s(X) \in I_b\}g(X)]. \quad (2.1)$$

This quantity is the signed calibration error contributed by group g inside score bin b . If $\theta_{b,g}^Q > 0$, then the model is underconfident on that subgroup-bin cell on average. If $\theta_{b,g}^Q < 0$, then it is overconfident. We use this moment because it is mathematically simple and still captures the same intuition as binned calibration.

Definition 1 (Groupwise binned reliability). For a subgroup $g \in \mathcal{G}$, define

$$\text{Cal}_{Q,\Pi}(g) = \sum_{b=1}^B |\theta_{b,g}^Q|. \quad (2.2)$$

The aggregate population is represented by the constant indicator $\mathbf{1}(x) = 1$. Thus $\text{Cal}_{Q,\Pi}(\mathbf{1})$ is the aggregate binned reliability discrepancy, while

$$\text{WCE}_{Q,\Pi}(\mathcal{G}) = \max_{g \in \mathcal{G}} \text{Cal}_{Q,\Pi}(g) \quad (2.3)$$

is the worst-group reliability discrepancy.

Definition 2 (Hidden-failure gap). The hidden-failure gap is

$$\text{HFG}_{Q,\Pi}(\mathcal{G}) = \text{WCE}_{Q,\Pi}(\mathcal{G}) - \text{Cal}_{Q,\Pi}(\mathbf{1}). \quad (2.4)$$

Large values of $\text{HFG}_{Q,\Pi}(\mathcal{G})$ mean that the overall metric is hiding a large subgroup failure. In the experiments we also report standard ECE-style metrics, which are easier to read in practice but serve the same purpose.

2.3 Weighted estimators

If oracle weights were available, a natural estimator of [Equation \(2.1\)](#) would be

$$\hat{\theta}_{b,g} = \frac{1}{n} \sum_{i=1}^n w(X_i)(Y_i - s(X_i))\mathbf{1}\{s(X_i) \in I_b\}g(X_i). \quad (2.5)$$

This is just a weighted average over source samples. Intuitively, a source point gets a large weight if that kind of point is common in the target population and rare in the source population. For

example, if a certain patient profile is about twice as common after deployment, then source samples with that profile should receive a weight near 2.

With estimated weights $\hat{w}$, we replace w by $\hat{w}$. This gives empirical estimators of $\text{Cal}_{Q,\Pi}(g)$ and $\text{WCE}_{Q,\Pi}(\mathcal{G})$ by plugging in $\hat{\theta}_{b,g}$.

The same idea extends beyond calibration. Let $u(X)$ be a selection score, for example confidence, and let

$$\mathcal{A}_\tau = \{x : u(x) \geq \tau\}$$

be the set of accepted examples. Then weighted transport also applies to accepted-set risk and accepted-set coverage. This matters when we interpret the selective-prediction experiments in [Section 4](#).

3 Theory

3.1 Exact transport under covariate shift

Proposition 1 (Transport identity). *Under [assumptions 1 and 2](#), for every measurable function $h : \mathcal{X} \times \{0, 1\} \rightarrow \mathbb{R}$ such that $\mathbb{E}_Q[|h(X, Y)|] < \infty$,*

$$\mathbb{E}_Q[h(X, Y)] = \mathbb{E}_P[w(X)h(X, Y)].$$

In particular, for every bin b and subgroup $g \in \mathcal{G}$,

$$\theta_{b,g}^Q = \mathbb{E}_P[w(X)(Y - s(X))\mathbf{1}\{s(X) \in I_b\}g(X)].$$

Remark 1. [Proposition 1](#) is the key identity in the paper. It says that if we want a target average, we can estimate it with a weighted source average. After that, the problem is no longer conceptual; it is statistical. We must estimate the weights, control their variability, and make sure the effective sample size does not collapse.

3.2 Uniform estimation over bins and groups

We now state a finite-family estimation result. Let $G = |\mathcal{G}|$ be the number of subgroups and assume the score partition Π is fixed in advance.

Assumption 3 (Bounded scores and weights). *Almost surely under P , we have $0 \leq s(X) \leq 1$, $Y \in \{0, 1\}$, and $0 \leq w(X) \leq W_{\max}$.*

Theorem 1 (Uniform control of weighted subgroup residuals). *Assume [assumptions 1 to 3](#). Let $\hat{w}$ be any estimated weight function and define*

$$\hat{\theta}_{b,g} = \frac{1}{n} \sum_{i=1}^n \hat{w}(X_i)(Y_i - s(X_i))\mathbf{1}\{s(X_i) \in I_b\}g(X_i).$$

Then there exists a universal constant $C > 0$ such that with probability at least $1 - \delta$,

$$\max_{1 \leq b \leq B, g \in \mathcal{G}} |\hat{\theta}_{b,g} - \theta_{b,g}^Q| \leq CW_{\max} \sqrt{\frac{\log(2BG/\delta)}{n}} + \mathbb{E}_P[|\hat{w}(X) - w(X)|]. \quad (3.1)$$

The bound has two pieces. The first term is the usual sampling term. It gets smaller as n grows. The second term is the weight-estimation term. It is small only if $\hat{w}$ is close to the true density ratio w . In other words, even with many source samples, the estimate can still be poor if the weights are inaccurate.

Proof sketch. Write

$$\hat{\theta}_{b,g} - \theta_{b,g}^Q = \underbrace{\frac{1}{n} \sum_{i=1}^n w(X_i) \psi_{b,g}(X_i, Y_i) - \mathbb{E}_P[w(X) \psi_{b,g}(X, Y)]}_{\text{sampling term}} + \underbrace{\frac{1}{n} \sum_{i=1}^n (\hat{w}(X_i) - w(X_i)) \psi_{b,g}(X_i, Y_i)}_{\text{weight error}},$$

where $\psi_{b,g}(x, y) = (y - s(x)) \mathbf{1}\{s(x) \in I_b\} g(x)$. Since $|y - s(x)| \leq 1$, the functions $w \psi_{b,g}$ are uniformly bounded by $W_{\max}$. A union bound over the finite family of BG functions gives the first term. The second term is bounded by $n^{-1} \sum_i |\hat{w}(X_i) - w(X_i)|$ and then by its expectation. A sharper version can replace the first term by a self-normalized empirical process bound. $\square$

Corollary 1 (Worst-group reliability estimation). *Under the assumptions of Theorem 1,*

$$\left| \widehat{\text{WCE}}_{\Pi}(\mathcal{G}) - \text{WCE}_{Q,\Pi}(\mathcal{G}) \right| \leq B \max_{1 \leq b \leq B, g \in \mathcal{G}} \left| \hat{\theta}_{b,g} - \theta_{b,g}^Q \right|,$$

where $\widehat{\text{WCE}}_{\Pi}(\mathcal{G}) = \max_{g \in \mathcal{G}} \sum_{b=1}^B |\hat{\theta}_{b,g}|$.

Remark 2 (Effective sample size). The sampling term in Equation (3.1) is often better interpreted through the empirical effective sample size

$$n_{\text{eff}} = \frac{(\sum_{i=1}^n \hat{w}_i)^2}{\sum_{i=1}^n \hat{w}_i^2}.$$

When the weights are fairly even, n_{eff} is close to n . When a few samples receive very large weights, n_{eff} can become much smaller than n . For example, one may start with 500 source points but end up with an effective sample size closer to 80. This is exactly the regime in which subgroup auditing becomes unstable first.

3.3 Why aggregate reliability can be non-diagnostic

Proposition 2 (Hidden failures can be invisible in aggregate). *There exist distributions P, Q , a score function s , and a subgroup family $\mathcal{G}$ such that aggregate calibration is perfect while every subgroup is badly calibrated. In particular, there exists a construction with*

$$\text{Cal}_{Q,\Pi}(\mathbf{1}) = 0 \quad \text{and} \quad \text{WCE}_{Q,\Pi}(\mathcal{G}) > 0.$$

Construction. Consider two groups g_A and g_B with equal target mass $1/2$. Let $Y = 0$ almost surely on group A and $Y = 1$ almost surely on group B . Let the score be constant: $s(X) = 1/2$ for all X . With a single score bin, the aggregate residual is

$$\mathbb{E}_Q[Y - s(X)] = \frac{1}{2}(0 - 1/2) + \frac{1}{2}(1 - 1/2) = 0,$$

so aggregate calibration is perfect. But each subgroup has conditional calibration gap $1/2$, and under the moment-based definition Equation (2.2), each subgroup contributes absolute residual mass $1/4$. Hence the aggregate diagnostic is exactly zero while subgroup diagnostics are strictly positive. $\square$

Remark 3. [Proposition 2](#) shows the basic failure mode of average metrics. Group A is always overconfident, group B is always underconfident, and the two errors cancel when we average them together. This is why a small aggregate ECE does not guarantee that every subgroup is safe.

3.4 Selective deployment as a transported subgroup problem

The same logic applies to abstention. Let $\ell(s(X), Y)$ be a bounded loss, for example 0–1 loss or log loss. Let $u : \mathcal{X} \rightarrow \mathbb{R}$ be a selection score, for example model confidence, and let

$$\mathcal{A}_\tau = \{x : u(x) \geq \tau\}$$

be the accepted set at threshold τ . For any subgroup g , the target selective-risk numerator is

$$\mathbb{E}_Q[\ell(s(X), Y)\mathbf{1}\{X \in \mathcal{A}_\tau\}g(X)],$$

which again equals a weighted source expectation under [Proposition 1](#). So coverage, selective risk, and subgroup rejection gaps are governed by the same overlap and effective-sample-size issues as weighted reliability estimation. This helps explain why selective classification can improve accepted-set averages while worsening subgroup disparities ([Geifman and El-Yaniv, 2017](#); [Jones et al., 2020](#)).

4 Preliminary Experiments

4.1 Experimental protocol

The current draft uses the repository’s existing synthetic benchmark suite as a controlled pilot study. The runs are generated from `paper_assets/configs/publication_suite.yaml` and cover three scenario families:

- **Covariate-shift sweep:** varying shift severity in a binary tabular classification task.
- **Subgroup degradation:** hidden failure modes concentrated in minority or masked subgroups.
- **Selective shift:** abstention-based deployment policies under increasing shift.

Each scenario currently uses only two seeds in the committed artifact suite, so the results below are best read as a serious pilot rather than a final statistical study. They are still useful because they match the main objects of the theory: effective sample size, target calibration, worst-group gaps, and selective risk.

To make the scenarios concrete:

- in the covariate-shift sweep, the same prediction task is kept fixed while the target covariates are moved farther away from the source covariates;
- in the subgroup benchmarks, failure is concentrated in one subgroup even when overall metrics remain moderate;
- in the selective benchmark, the model may reject low-confidence examples and we measure the risk on the accepted set.

Table 1: Preliminary benchmark summary extracted from the committed publication suite. These numbers are not the final experimental campaign; they are the current strongest evidence already present in the repository.

Scenario	Setting	Observation
Covariate shift	severity 0.20	effective sample size ≈ 179.8 ; raw target ECE ≈ 0.075 ; recalibrated target ECE ≈ 0.064
Covariate shift	severity 1.40	effective sample size ≈ 60.9 ; raw target ECE ≈ 0.064 ; recalibrated target ECE ≈ 0.057
Masked subgroup shift	audit baseline	worst-group accuracy gap ≈ 0.137 ; worst-group ECE gap ≈ 0.182
Minority subgroup degradation	audit baseline	worst-group accuracy gap ≈ 0.140 ; worst-group ECE gap ≈ 0.217
Selective shift	severity 0.40	weighted confidence abstention keeps coverage ≈ 0.707 with target risk reduction ≈ 0.079 and selective ECE ≈ 0.070
Selective shift	severity 1.00	weighted confidence abstention keeps coverage ≈ 0.759 with target risk reduction ≈ 0.061 and selective ECE ≈ 0.084

4.2 Key empirical patterns

Effective sample size degrades as shift severity increases. The covariate-shift benchmark exhibits a clear collapse in effective sample size, from roughly 180 at severity 0.20 to roughly 61 at severity 1.40. This matches [Theorem 1](#): even if weighted transport is valid in principle, the estimation problem becomes harder when a few source points carry most of the weight.

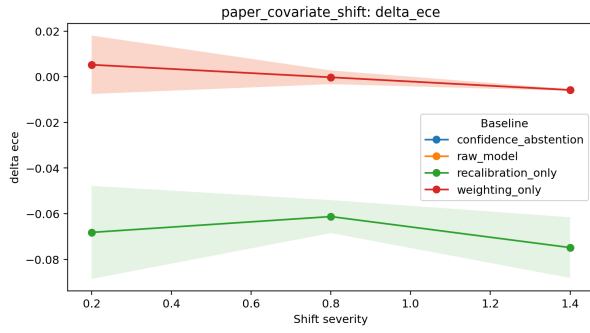
Aggregate target error can look moderate while worst-group gaps are large. In the hidden subgroup scenarios, the subgroup-audit baseline reports worst-group ECE gaps of about 0.182 and 0.217, together with worst-group accuracy gaps around 0.14. In simple terms, the average target metric looks much better than the worst affected subgroup. This is exactly the hidden-failure phenomenon formalized earlier.

Selective policies improve accepted-set averages but require subgroup-aware interpretation. In the selective-shift pilot, weighted and recalibrated abstention policies retain roughly 71% to 76% coverage while reducing target selective risk. The weighted policy also achieves better accepted-set calibration than the recalibrated confidence baseline at both severities in the committed suite. The lesson is simple: abstention can help, but it should still be evaluated with subgroup-aware target metrics.

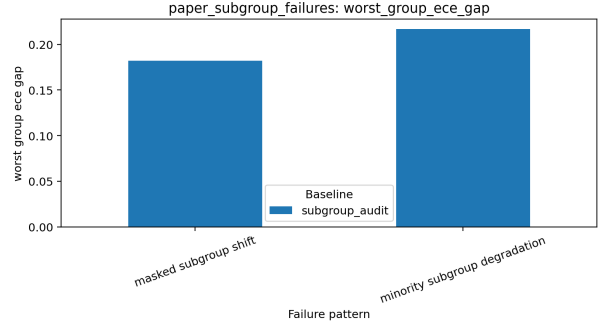
4.3 How the final paper should extend these experiments

The current pilot is enough to support the theoretical framing, but not enough for a finished preprint. The final experimental section should strengthen the manuscript in four ways:

1. increase the number of seeds so uncertainty intervals become meaningful;



(a) Calibration change across covariate-shift severity.



(b) Worst-group calibration gap in hidden-failure scenarios.

Figure 1: Two pilot figures from the repository’s benchmark suite. The left panel shows how a calibration-oriented metric changes as shift severity increases. The right panel shows that worst-group ECE gaps stay large in subgroup-failure scenarios.

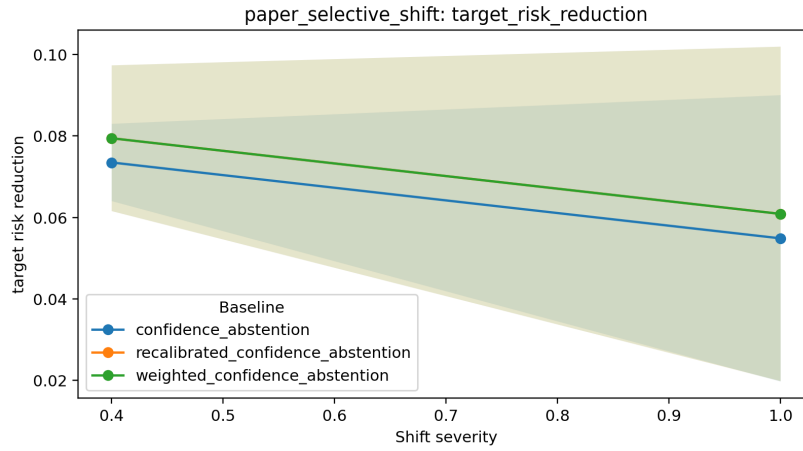


Figure 2: Target risk reduction in the selective-shift benchmark. Even when abstention improves average accepted-set risk, it should still be read together with subgroup rejection behavior and accepted-set calibration.

2. add ablations on density-ratio quality and explicit overlap violation;
3. vary subgroup family complexity, from single features to intersectional slices;
4. include at least one non-synthetic or semi-real tabular benchmark where subgroup prevalence shift is operationally plausible.

These extensions would turn the current manuscript from a strong theory draft with executable evidence into a much more convincing empirical paper.

5 Discussion

The main message of this paper is simple: reliability under deployment shift should not be summarized by a single average score. Under covariate shift, target reliability can often be estimated from

weighted source data, but this is only useful if we also ask which subgroups are being audited and how much effective sample size remains after weighting.

This perspective clarifies several common points of confusion.

- **Weighted source diagnostics and target diagnostics are not the same thing.** The weighted source estimate is only valid under the transport assumptions, and it becomes unstable when overlap is weak.
- **A good average can hide a bad subgroup.** The hidden-failure construction makes this precise in the simplest possible way.
- **Selective deployment does not remove the problem.** Once abstention is written in target-risk form, it inherits the same weighting and overlap issues as calibration estimation.

There are several directions that would make the final paper more technically ambitious.

1. Replace the finite-family subgroup analysis with VC- or Rademacher-style complexity control for richer slice families.
2. Derive self-normalized or semiparametric bounds that sharpen the role of effective sample size.
3. Move from estimation error bounds to inference, for example confidence intervals on worst-group reliability or hidden-failure gaps.
4. Extend the covariate-shift story to label shift, concept drift, and mixed shift regimes, where weighted transport is no longer exact.

The current repository provides a useful computational scaffold for these questions, but the scientific point is independent of the software itself. The deeper claim is conceptual: deployment robustness should be analyzed as a weighted subgroup reliability problem with explicit statistical limits, not as a single post-hoc calibration number.

References

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A Appendix

A.1 Proof of Proposition 1

For any integrable measurable function h ,

$$\mathbb{E}_P[w(X)h(X, Y)] = \int h(x, y) \frac{dQ_X}{dP_X}(x) dP(x, y).$$

Since under [assumption 1](#) the conditional law of Y given $X = x$ is the same under P and Q , we may write

$$dP(x, y) = dP_X(x) dP(y | x) \quad \text{and} \quad dQ(x, y) = dQ_X(x) dQ(y | x),$$

with $dP(y | x) = dQ(y | x)$. Therefore

$$\frac{dQ_X}{dP_X}(x) dP_X(x) dP(y | x) = dQ_X(x) dQ(y | x) = dQ(x, y),$$

which yields

$$\mathbb{E}_P[w(X)h(X, Y)] = \mathbb{E}_Q[h(X, Y)].$$

Choosing

$$h(x, y) = (y - s(x))\mathbf{1}\{s(x) \in I_b\}g(x)$$

gives the stated formula for $\theta_{b,g}^Q$.

A.2 Further explanation of the hidden-failure construction

The construction in [Proposition 2](#) is intentionally simple because the phenomenon itself is simple. If subgroup-specific errors have opposite signs, they can cancel in the aggregate. In the two-group example:

- group A is maximally overconfident relative to the constant score $1/2$;
- group B is maximally underconfident relative to the same score;
- the aggregate average is perfectly calibrated because the two signed residuals cancel.

This cancellation logic survives in much richer settings. The synthetic subgroup benchmarks in the repository are essentially structured versions of the same idea, now combined with prevalence shift and score heterogeneity.

A.3 Experimental provenance

The preliminary results cited in the main text come from the committed benchmark artifacts:

- `../paper_assets/generated/publication_suite/paper_covariate_shift/`
- `../paper_assets/generated/publication_suite/paper_subgroup_failures/`

- `../paper_assets/generated/publication_suite/paper_selective_shift/`

The pilot suite is driven by:

- `../paper_assets/configs/publication_suite.yaml`

For the finished paper, the experimental section should be regenerated with a larger seed budget and explicit ablations on overlap and density-ratio quality.